Lesson notes · for you

Relationships

Data science · Level 2 · Lesson 3

Time
30 min

Materials
A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Draw and read a scatter plot.
- Read a correlation r.
- Ask what else could explain a pattern.

Activity: A class scatter chart

1. Warm-up (5 min): Guess whether taller pupils have longer arms, then say how you could check.
2. Plot (10 min): Measure with a partner and add your dot to the axes on the board.
3. Discuss (10 min): Decide how strong the pattern is, then find another explanation for two claims.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- How strong was our pattern?
- What links ice cream and sunburn?
- How could we test a cause fairly?

Check yourself

- Sketch a scatter plot with r close to -0.9.

Try it this week
Measure the height and arm span of five people at home and plot them on a scatter chart. Is there a pattern, and does one cause the other?

Worksheet

Relationships

Data science · Level 2 · Lesson 3

Name _____ Class _____ Date _____

1 **More books at home goes with higher marks (r = 0.6). Does that prove books cause higher marks?**

◯ No: something else may explain both ◯ Yes ◯ Only if r is 1

2 **Match each pattern to its r.**

Dots close to a rising line	• • • •	About 0
A shapeless cloud	• • • •	About -0.9
Dots close to a falling line	• • • •	About 0.9

3 **Finish the program so it prints r between study hours and marks, to 2 decimal places.**

```
x = [1, 2, 3, 4, 5]
y = [8, 9, 11, 12, 15]
mx, my = 3, 11
sxy = sxx = syy = 0
for i in range(5):
    sxy += (x[i] - mx) * (y[i] - my)
print(sxy)
```

Write your program:

4 **Because ice cream sales and sunburn rise together, ice cream causes sunburn.**

◯ True ◯ False

5 **Teens who sleep more get better marks. What else could explain it? Choose all that apply.**

☐ A calm home ☐ Less screen time late at night ☐ Their shoe size ☐ Good health

✎ **Which "fact" have you heard that might be a correlation, not a cause?**

Check your answers

Relationships

1 **No: something else may explain both**

Even a strong correlation is not proof of cause.

2 **Dots close to a rising line — About 0.9**

A shapeless cloud — About 0

Dots close to a falling line — About −0.9

The sign gives the direction, the size how close to a line.

3

```
x = [1, 2, 3, 4, 5]
y = [8, 9, 11, 12, 15]
mx, my = 3, 11
sxy = sxx = syy = 0
for i in range(5):
    sxy += (x[i] - mx) * (y[i] - my)
    sxx += (x[i] - mx) ** 2
    syy += (y[i] - my) ** 2
print(round(sxy / (sxx * syy) ** 0.5, 2))
```

$r = 17 \div \sqrt{(10 \times 30)} \approx 0.98$: a very strong positive relationship for these five pupils, but it still doesn't prove studying alone causes the marks.

4 **False**

Sunny weather causes both. Correlation is not causation.

5 **Less screen time late at night; Good health; A calm home**

Each of the first three could raise both sleep and marks.